PROJECT AND TEAM INFORMATION

Project Title

(Try to choose a catchy title. Max 20 words).

Smart Traffic Flow Optimization and Emergency Preemption Simulation Engine in C++

Student / Team Information

Team Name: Traffic Flow Optimization Team *Team # DSCPP-321*	*FlowOptimizers*
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PROPOSAL DESCRIPTION (10 pts)

Motivation (1 pt)

(Describe the problem you want to solve and why it is important. Max 300 words).

Conventional traffic systems rely heavily on static, fixed-time signals that cannot adapt to real-time density fluctuations, causing gridlock, excessive vehicle idling, and increased emissions. Furthermore, traditional systems fail to prioritize emergency vehicles (ambulances, fire engines), leading to life-threatening delays in dense traffic corridors. Solving this problem through automated, data-driven signal adjustments and intelligent dynamic rerouting minimizes average commuter wait times, flushes bottlenecks before gridlock occurs, and establishes rapid, zero-delay emergency corridors across urban grids.

State of the Art / Current solution (1 pt)

(Describe how the problem is solved today (if it is). Max 200 words).

Existing urban setups primarily rely on pre-programmed cyclic timers or isolated sensor loops. Advanced municipal systems like SCATS and SCOOT adjust signal splits based on regional saturation, while Indian cities use ITMS with ANPR cameras. However, legacy deployments often lack automated priority preemption for emergency vehicles and do not perform graph-wide shortest-path rerouting across adjacent intersections when localized bottlenecks exceed critical capacity.

Project Goals and Milestones (2 pts)

(Describe the project general goals. Include initial milestones as well any other milestones. Max 300 words).

The primary goal is to build a high-performance C++ discrete-time traffic simulation engine and interactive 2D GUI to validate adaptive signal control and priority preemption against traditional static systems.
Milestone 1: Implement OOP vehicle hierarchy and FIFO/Min-Heap queue structures.
Milestone 2: Construct the Adjacency List graph representing intersections and dynamic Dijkstra rerouting.
Milestone 3: Build adaptive SCATS-inspired green-split algorithms and emergency preemption logic.
Milestone 4: Develop the interactive 2D SFML/Raylib UI with real-time benchmarking metrics.

Project Approach (3 pts)

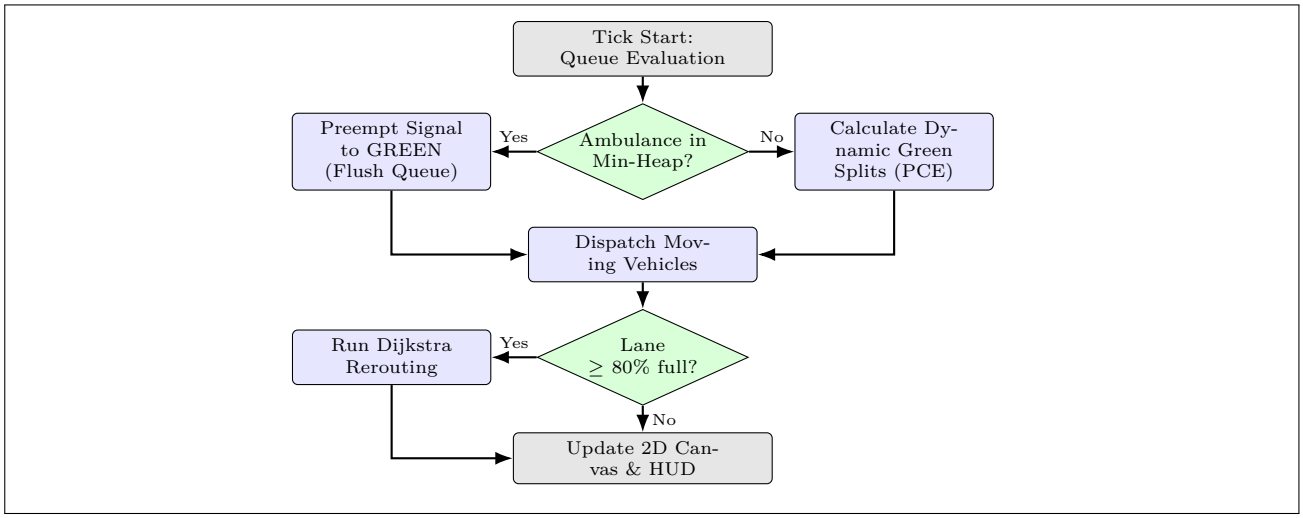
(Describe how you plan to articulate and design a solution. Including platforms and technologies that you will use. Max 300 words).

We will implement a modular C++ (C++17/20) engine utilizing OOP and core DSA principles:

- **Data Structures:** Graphs (Adjacency Lists) for road networks, FIFO Queues (`std::queue`) for lane vehicle queues, and Min-Heap Priority Queues (`std::priority_queue`) for emergency preemption.
- **Algorithms:** Dynamic green split calculation based on Passenger Car Equivalent (PCE) lane density, and Dijkstra's algorithm for dynamic rerouting when lane saturation $\geq 80\%$.
- **Platform/UI:** Built natively in C++ using SFML/Raylib for a 2D interactive top-down simulation featuring a side-by-side toggle (Static vs. Dynamic).

System Architecture (High Level Diagram)(2 pts)

(Provide an overview of the system, identifying its main components and interfaces in the form of a diagram using a tool of your choice).



Project Outcome / Deliverables (1 pts)

(Describe what are the outcomes / deliverables of the project. Max 200 words).

The primary deliverable is a functional C++ interactive simulation executable featuring:

- A dynamic signal controller engine that reduces average vehicle wait time.
- A Priority Queue module ensuring zero-delay emergency preemption corridors.
- A 2D visualization dashboard displaying live metrics (throughput, average wait time reduction, congestion index) with a toggle switch between Static and Smart modes.

Assumptions

(Describe the assumptions (if any) you are making to solve the problem. Max 100 words)

Lane queues and vehicle densities are measured by infrastructure sensors (cameras/induction loops). Heterogeneous traffic is modeled using standardized Passenger Car Equivalents (PCE). Emergency vehicles are authenticated and detected within a fixed approach radius.

References

(Provide a list of resources or references you utilised for the completion of this deliverable. You may provide links).

1. Lowrie, P. R. (1990). *SCATS: Sydney Coordinated Adaptive Traffic System.*
2. Cormen, T. H., et al. (2009). *Introduction to Algorithms* (Graph Algorithms & Heaps).
3. SFML Development Team. (2024). *Simple and Fast Multimedia Library Documentation.* https://www.sfml-dev.org